

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Nasarawa State, Nigeria

LAB REPORT

Lab 04: Introduction to Supervisory Control and Data Acquisition (SCADA)

Item	Details
Lab Title	Lab 04: Introduction to Supervisory Control and Data Acquisition (SCADA)
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1. Executive Summary

This laboratory introduced SCADA (Supervisory Control and Data Acquisition). SCADA is the system that operators use to watch and supervise an industrial process from a control room. The practical used two ICDFA virtual machines: a PLC machine running OpenPLC Runtime, and a SCADA machine running the FUXA HMI. The process being supervised was an automated liquid-storage tank.

I started FUXA, loaded the supplied tank-fill HMI project (TankFillTest.json), and connected it to the PLC at 192.168.1.10, port 502, using Modbus TCP. On the PLC machine I uploaded and compiled the TankFillSim program, started it, and confirmed that the runtime status was Running. Back in FUXA, I ran the HMI and observed the tank level, the valve indication and the process trend during automatic filling and draining.

The report explains how SCADA gives operators central visibility through displays, alarms, event records and historical data, and how the PLC does the real control while SCADA only supervises. All work stayed inside the isolated laboratory.

2. Lab Objectives

By the end of this lab, I should be able to:

- Explain the purpose and architecture of a SCADA system.
- Explain the roles of PLCs, RTUs, SCADA servers and HMIs, and the purpose of alarms, event recording and historization.
- Start FUXA on the SCADA VM and load a prepared HMI project.
- Connect the FUXA HMI to OpenPLC Runtime over Modbus TCP in the isolated laboratory.
- Start the OpenPLC tank-fill simulation, then operate and observe the HMI.
- Interpret tank level, valve state and process trends, and explain how SCADA gives operators central visibility.

3. Tools and Resources Used

Tool / Resource	What it was used for
PLC VM (192.168.1.10)	Runs OpenPLC Runtime, which is the PLC that controls the tank simulation.
SCADA VM (192.168.2.10)	Runs FUXA, which is the SCADA server and HMI used by the operator.
Virtualisation platform: _____ (version _____)	Used to run both VMs on the isolated laboratory network. Write the name and version you used.
OpenPLC Runtime (web page: localhost:8080)	Used to upload, compile, start and stop the TankFillSim program.
FUXA (editor: 127.0.0.1:1881/editor), version _____	Used to load the HMI project, set the PLC connection and run the HMI.

Tool / Resource	What it was used for
Firefox browser	Used to open the FUXA editor and the OpenPLC Runtime web page.
TankFillTest.json and TankFillSim.st	The supplied HMI project file and the supplied PLC program.
Modbus TCP, port 502	The industrial protocol used between FUXA and OpenPLC.
Screenshot tool	Used to save evidence at each step.

4. Methodology (Step-by-Step Procedure)

I followed the 12 steps of the lab brief in order. A screenshot was captured at each important step. The screenshots are numbered Figures 1 to 12. Passwords are not written in this report.

4.2 The Steps

Step 1: Start both VMs. I imported and started the IC DFA PLC VM and SCADA VM using the isolated network configuration given by my instructor. I created the **LAB04-START** snapshot for each VM (if it was not already available). I did not use Bridged networking.

5. Evidence and Analysis

5.1 Evidence 1: Access to the VMs, FUXA Launch, Project Load and Connection

Question: Show access to the PLC and SCADA VMs, the launch of FUXA, the loading of the tank-fill HMI project, and the connection to the authorised PLC address (Figures 1 to 5).

Part A: The two machines

Machine	Address	Software	Role
PLC VM	192.168.1.10	OpenPLC Runtime	Runs the control program that fills and drains the tank.
SCADA VM	192.168.2.10	FUXA	Collects data from the PLC and shows it to the operator.

Part B: What each figure proves

Figure	What it proves
Figure 1	Both the PLC VM and the SCADA VM were started in the authorised isolated laboratory, from clean snapshots.
Figure 2	The FUXA SCADA service was started on the SCADA VM with <code>sudo fuxa</code> .
Figure 3	The FUXA editor was reached in the browser at <code>127.0.0.1:1881/editor</code> .
Figure 4	The supplied HMI project <code>TankFillTest.json</code> was loaded.
Figure 5	The <code>openplc</code> device was pointed to the authorised PLC address, <code>192.168.1.10</code> , on port <code>502</code> .

Part C: Why the connection is set to **192.168.1.10:502**

FUXA needs to know where the PLC is. The address **192.168.1.10** is the PLC VM, and **502** is the standard port for Modbus TCP. In this connection, FUXA is the client that asks and OpenPLC Runtime is the server that answers. FUXA asks the PLC for the values of the process, such as the tank level and the valve state, again and again, and shows them on the HMI.

5.2 Evidence 2: OpenPLC Runtime in the Running State

Question: Show the startup of the OpenPLC tank-fill simulation (Figures 6 to 8).

Figure	What it shows and what it means
Figure 6	I signed in to OpenPLC Runtime on the PLC VM at <code>localhost:8080</code> . The word <code>localhost</code> means the PLC VM itself.
Figure 7	<code>TankFillSim.st</code> was uploaded and named Tank Fill Sim. It was compiled, which changes the program text into code the PLC runtime can execute.
Figure 8	The Dashboard status shows Running. This means the PLC is now executing the tank-fill program and is ready to answer FUXA.

Answer: Figure 8 is the proof that the PLC was in the Running state. Without a running PLC, FUXA would have no live values to show, so this step had to come before running the HMI.

5.3 Evidence 3: FUXA HMI in Operation

Question: Show the FUXA HMI operating while the tank fills and drains automatically (Figures 9 to 11), and record what was observed (Step 10).

Part A: Observation worksheet

Write what you actually saw on your screen. Do not copy numbers from anywhere else.

Item observed
Lowest tank level seen
Highest tank level seen
Valve indication when OPEN (how it looks or what it says)
Valve indication when CLOSED
Level at the moment the valve opened
Level at the moment the valve closed
Time for one full fill and drain cycle (about)
Trend shape (describe the line going up and down)
Alarm / event / history screen available in the project? (yes / no)

Part B: How the automatic process works (simple explanation)

The PLC program TankFillSim controls the tank by itself. When the level is low, it opens the fill valve and the level rises. When the level is high, it closes the valve and the level falls as the liquid drains. Then the cycle repeats. FUXA does not control this. It only reads the values and shows them.

On the HMI I could see this cycle as follows:

Stage	What the operator sees on the HMI
Filling	Valve indication shows OPEN. The tank level rises. The trend line goes up.
Level reaches its upper limit	Valve indication changes to CLOSED. The trend line stops rising.
Draining	Valve indication stays CLOSED. The tank level falls. The trend line goes down.
Level reaches its lower limit	Valve indication changes to OPEN again and a new fill starts.

Check on your screen: confirm that your HMI behaves like this and change the table if your project shows something different.

5.4 Evidence 4: Analysis of the Displays and SCADA Functions

Question: Explain the displayed tank level, valve status, process trend, alarming, event recording and historization, and how they help operators.

Function	What it is	How it helps the operator
Tank level	The measured height of liquid in the tank, sent by the PLC and shown as a number and / or a moving tank picture.	The operator can see at a glance whether the tank is nearly empty, normal or nearly full.
Valve status	A symbol or text showing whether the fill valve is OPEN or CLOSED.	It explains why the level is rising or falling and shows that the PLC is doing what it should.
Process trend	A graph of a value against time, such as the tank level over the last minutes.	It shows direction and speed. The operator can see whether the level is rising, falling or stuck, and spot problems early.
Alarming	A warning that appears when a value goes outside its safe limit, for example a very high level.	It draws attention to a problem at once, so the operator can act before damage or overflow.
Event recording	A time-stamped log of things that happened: alarm raised, valve changed state, PLC stopped, operator command.	It answers "what happened and when?" and helps find the cause of a problem afterwards.
Historization	Continuous storage of process values over a long time (in a historian / database).	It allows reports, comparison with earlier days, and study of long-term problems.

Check on your screen: if your FUXA project does not show an alarm, event or history screen, write "not shown in this project" next to it in your notes. The meaning and the benefit above stay the same.

Part B: How SCADA supports operators

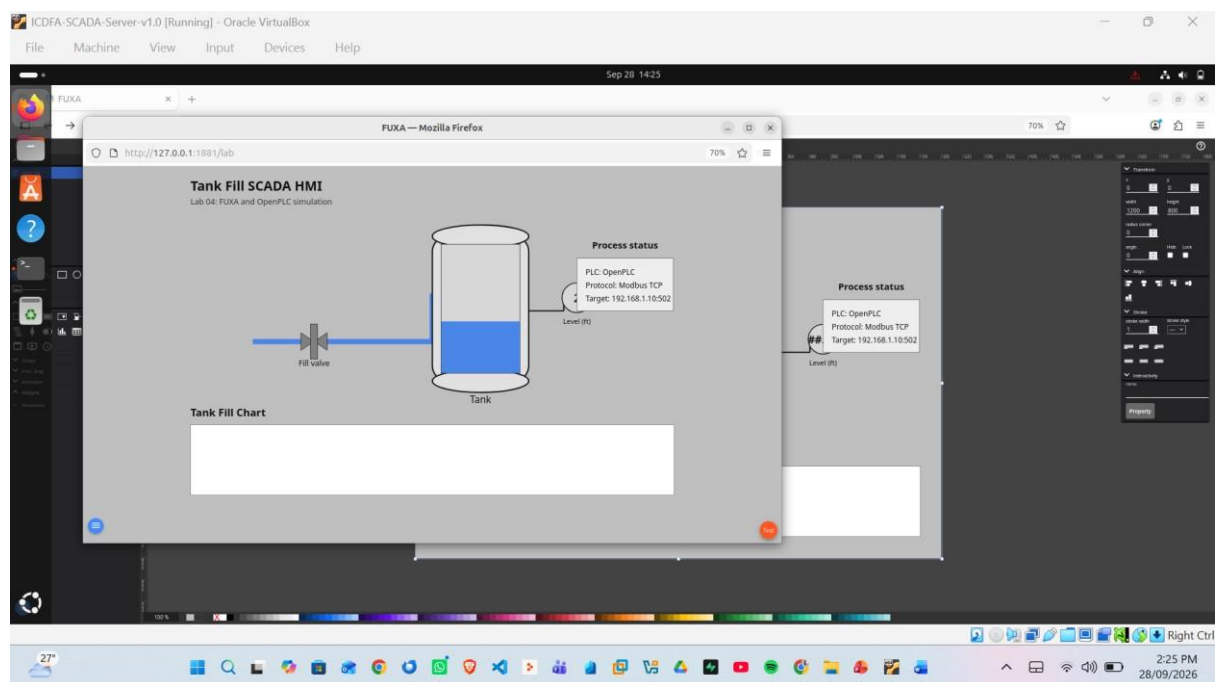
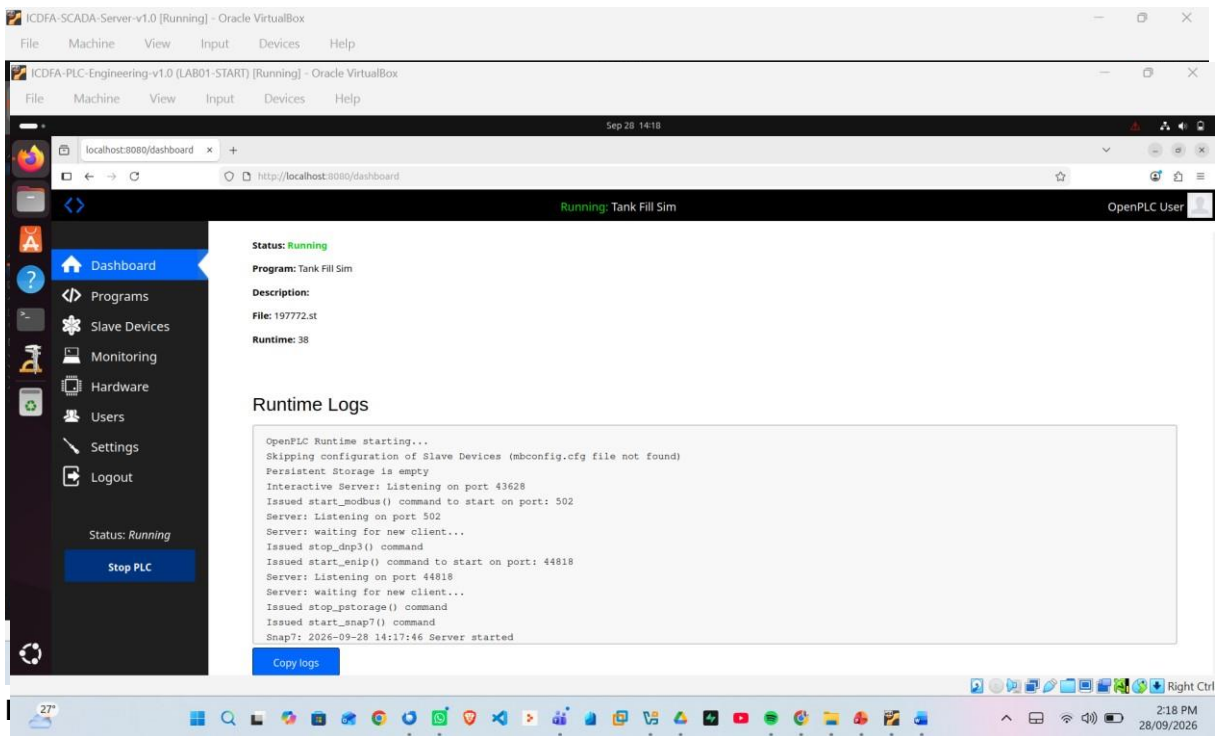
Without SCADA, an operator would have to walk to the tank and read a gauge. With SCADA, the operator sits in one place and sees the level, the valve state and the trend of the whole process on one screen, gets an alarm when something is wrong, and can look back at the record. This is called **central visibility**. It saves time, lets problems be found earlier, and makes decisions safer.

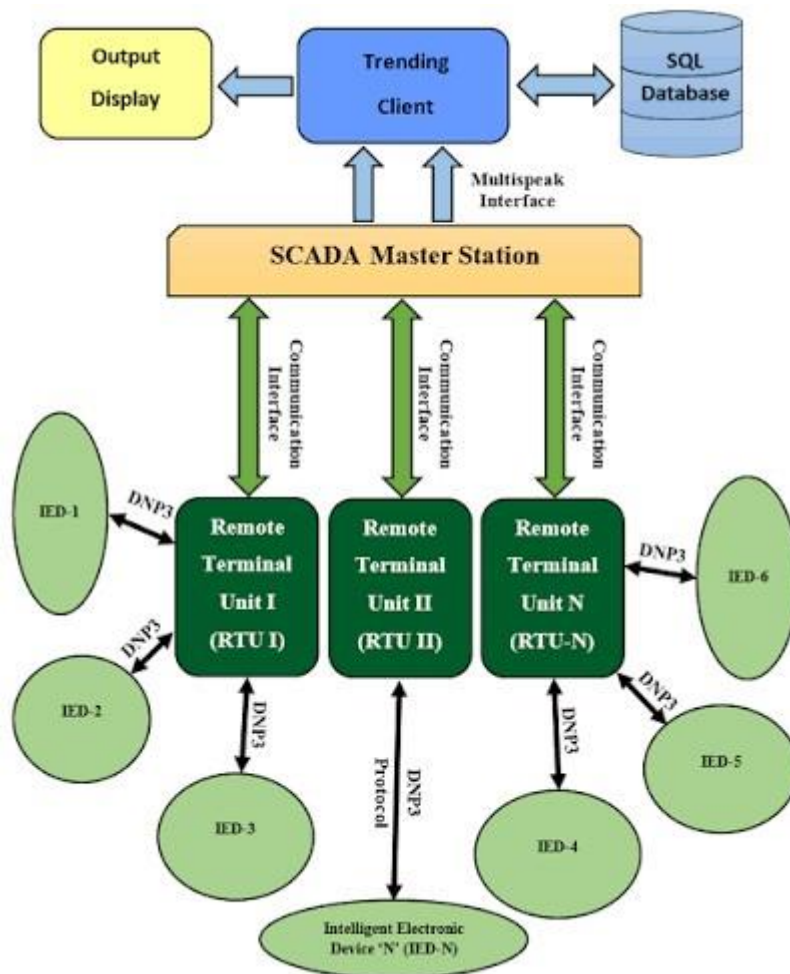
Part C: PLC, RTU, SCADA server and HMI

Component	Role	In this lab
PLC	Reads the field inputs, runs the control logic and drives the outputs.	OpenPLC Runtime on the PLC VM.
RTU (Remote Terminal Unit)	Similar to a PLC but built for remote sites such as a pumping station. It collects data and sends it back to the SCADA centre, often over radio or cellular links.	Not used in this lab. The PLC does its job here.
SCADA server	Collects data from PLCs / RTUs, applies alarms, records events and stores history.	FUXA on the SCADA VM.
HMI	The screen the operator looks at and uses.	The FUXA tank-fill HMI in the browser.

5.5 Evidence 5: SCADA Architecture Diagram

Question: Draw a labelled SCADA architecture diagram: Field/Process > PLC > Industrial Protocol/Network > SCADA/HMI > Operator, and show where alarms, events and the historian belong (Step 11).





Key: ^ = information going up to the operator, v = commands going down to the process.

Where alarms, events and the historian belong

Function	Where it belongs	Simple meaning
Alarms	SCADA / HMI layer (the PLC provides the values)	Warn the operator when a value is outside its limit.
Event recording	SCADA / HMI layer	A time-stamped diary of what happened.

Function	Where it belongs	Simple meaning
Historian (historization)	SCADA / HMI layer, stored in a database	Stores values over time for trends and reports.

How to read the diagram: The tank, sensor and valve are the process. The PLC reads the sensor and controls the valve. The PLC value is carried over Modbus TCP to FUXA. FUXA shows it to the operator and also raises alarms, records events and stores history. The operator can then give commands, which travel down the same path.

5.6 Evidence 6: PLC Control Logic versus SCADA Supervision

The PLC and the SCADA/HMI layer do different jobs. The PLC holds the control logic. It reads the level sensor, decides whether the fill valve must be open or closed, and switches the output. It does this by itself many times every second, and it keeps controlling the tank even if the SCADA screen is closed or the network fails. Its purpose is to keep the process running safely and correctly in real time.

The HMI/SCADA layer does not control the process moment by moment. It supervises. FUXA reads values from the PLC over Modbus TCP and shows them to the operator: the tank level, the valve state, trends and alarms. It records events and stores historical data so the operator can see what happened earlier. The operator can also give supervisory commands, but the PLC still carries out the actual control.

In simple words, the PLC is the worker who operates the valve, and SCADA is the supervisor who watches the whole plant from the office. Keeping them separate makes the plant safer, because control continues even when supervision is lost, and it gives operators clear visibility of the process.

5.7 Knowledge Check

1. What is the primary purpose of an HMI in a SCADA system?

The HMI (Human-Machine Interface) is the screen through which the operator sees and understands the process. It shows values, states, trends and alarms in a clear graphical way and lets the operator give supervisory commands.

2. What is the difference between an alarm, an event record and historized process data?

An alarm is a warning shown at once when a value is abnormal, for example a very high tank level, and it needs the operator to act. An event record is a time-stamped log entry of something that happened, such as an alarm being raised, a valve changing state or an operator command. Historized process data is the continuous storage of values over time, such as the tank level every few seconds, used for trends, reports and analysis.

3. Why should a SCADA operator have context rather than only raw sensor values?

A raw number alone means little. The operator needs units, limits, the trend and the state of related equipment to know whether the value is normal, rising or falling, and dangerous or safe. Context allows fast and correct decisions and prevents misreading.

4. What role does the PLC play compared with the SCADA application?

The PLC does the real-time control: it reads inputs, runs the logic and drives the outputs, even without SCADA. The SCADA application supervises: it collects data from PLCs or RTUs, shows it on the HMI, raises alarms, records events, stores history and passes operator commands to the PLC.

7. Conclusion

This lab showed how a SCADA system works together with a PLC. The PLC ran the tank-fill program and controlled the valve, while FUXA read the values over Modbus TCP and gave the operator a clear view of the tank level, valve state and trend. I also learned that alarms, event recording and historization belong to the SCADA layer, and that they help operators notice problems, find causes and study the process over time. Because the PLC controls the process and SCADA supervises it, the process keeps running safely even if supervision is lost.

8. Recommendations

- Keep SCADA and PLC systems on isolated or separated networks. Never connect a training lab to production or organisational networks.
- Modbus TCP has no built-in login or encryption, so it should not be exposed to untrusted networks. Use network separation and firewalls to protect it.
- Change default passwords, do not reuse credentials, and give operators only the access they need.
- Design HMI screens to give context (units, limits, trends and alarms), not only raw numbers.
- Always stop the simulation cleanly and restore the clean snapshot after the lab.

9. References

International Cybersecurity and Digital Forensics Academy (ICDFA). (2026). Lab 04: Introduction to Supervisory Control and Data Acquisition (SCADA) (Student Edition). Intro to Critical Infrastructure for IT Learners. <https://www.icdfa.edu.ng>

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